Math 1200 Lecture 19

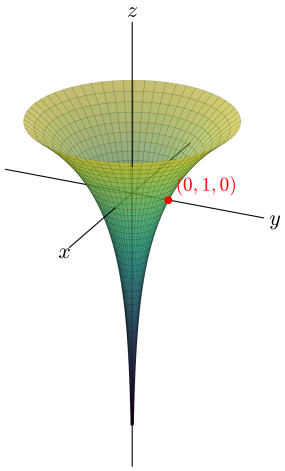
Nov 10 2026

Topics: 16.6: Tangent spaces, surface area Name:_____
Collaborators:_____

Group work 1 When the graph $z = f(y)$ is rotated around the z axis the surface of revolution is parametrized by

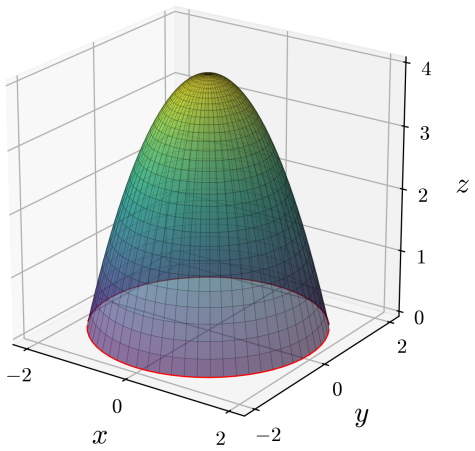
$$\vec{r}(u, v) = \langle f(u)\cos(v), f(u)\sin(v), u\rangle.$$

When $f(y) = e^y$ Find the tangent plane to the surface of revolution at the point $(u, v) = (0, \pi/2)$.[1]



[1] Make sure your answer matches your intuition from the picture

Group work 2



Consider the surface $\vec{r}(u, v)$ which is the part of the graph of $f(x, y) = 2 - x^2 - y^2$ above the xy-plane. Find its surface area.

Group work 3 Suppose $\vec{r}(u,v)$ parametrizes the graph S of a surface $z = f(x,y)$ in a region D so that $\vec{r}(u,v) = (u,v,f(u,v))$.

a) In general, show that $\vec{r}_u \times \vec{r}_v = \langle -f_x, -f_y, 1\rangle$

b) Using (a), conclude that the surface area of S is $A = \iint_D \sqrt{1+f_x^2+f_y^2}dA$.